

Linux Command Line Interface Notes

I. The Shell

What is the shell?

- A program that takes commands from keyboard and sends them to the operating system
- Terminal/Console programs launch shells for you
- Default shell: **bash** (Bourne Again shell)
- Other shells: ksh, zsh, tsch

Shell Prompt Format:

```
username@hostname:current_directory  
pete@icebox:/home/pete $
```

- `$` indicates normal user using Bash/Bourne/Korn shell
- Don't type the `$` when entering commands

Basic Command:

```
echo Hello World
```

2. Navigation Commands

pwd (Print Working Directory)

```
pwd
```

- Shows your current directory location
- Path starts from root directory (/)

cd (Change Directory)

```
cd /home/pete/Pictures    # Absolute path  
cd Hawaii                 # Relative path
```

Path Types:

- **Absolute path:** Starts from root (/) - e.g., /home/pete/Desktop
- **Relative path:** From current location - e.g., taxes/

Navigation Shortcuts:

- `.` - Current directory
- `..` - Parent directory (one level up)
- `~` - Home directory
- `-` - Previous directory

```
cd .      # Stay in current directory
cd ..     # Go up one level
cd ~      # Go to home directory
cd -      # Go to previous directory
```

ls (List Directories)

```
ls          # List current directory
ls /home/pete # List specific directory
ls -a       # Show all files (including hidden)
ls -l       # Long format with details
ls -la      # Combine flags
```

ls -l Output Details:

- File permissions
- Number of links
- Owner name
- Owner group
- File size
- Last modification timestamp
- File/directory name

3. File Operations

touch (Create Files)

```
touch mysuperduperfile
```

- Creates new empty files
- Updates timestamps on existing files

file (Check File Type)

```
file banana.jpg
```

- Shows description of file contents
- Linux filenames don't need to match file type

cat (Display File Contents)

```
cat dogfile birdfile
```

- Displays file contents
- Can combine multiple files
- Best for short content only

less (View Large Files)

```
less /home/pete/Documents/text1
```

Navigation in less:

- `q` - Quit
- Arrow keys/Page Up/Down - Navigate
- `g` - Go to beginning
- `G` - Go to end
- `/search` - Search for text
- `h` - Help

4. File Management

cp (Copy)

```
cp mycooolfile /home/pete/Documents/cooldocs
cp *.jpg /home/pete/Pictures          # Copy all .jpg files
cp -r Pumpkin/ /home/pete/Documents  # Copy directory recursively
cp -i mycooolfile /home/pete/Pictures # Interactive (prompt before
overwrite)
```

Wildcards:

- `*` - All characters or any string
- `?` - One character
- `[]` - Any character within brackets

mv (Move/Rename)

```
mv oldfile newfile           # Rename file
mv file2 /home/pete/Documents # Move file
mv file_1 file_2 /somedirectory # Move multiple files
mv directory1 directory2      # Rename directory
mv -i directory1 directory2   # Interactive
mv -b directory1 directory2   # Create backup
```

mkdir (Make Directory)

```
mkdir books paintings        # Create multiple directories
mkdir -p books/hemingway/favorites # Create subdirectories
```

rm (Remove)

```
rm file1                     # Remove file
rm -f file1                  # Force remove (no prompts)
rm -i file                   # Interactive
rm -r directory              # Remove directory recursively
rmdir directory              # Remove empty directory
```

⚠ **Warning:** Files removed with `rm` are permanently deleted!

5. Search and Help

find (Search for Files)

```
find /home -name puppies.jpg # Find by name
find /home -type d -name MyFolder # Find directory by name
```

File Types:

- `d` - Directory
- `f` - Regular file

help (Get Command Help)

```
help echo
echo --help
```

```
# Help for bash built-ins
# Help for other programs
```

man (Manual Pages)

```
man ls
```

- Detailed documentation for commands
- Built into most Linux systems

whatis (Brief Description)

```
whatis cat
```

- Provides brief description of commands
- Sourced from manual pages

6. Useful Features

history (Command History)

```
history
```

- Shows previous commands
- `↑` arrow - Previous command
- `!!` - Run last command
- `Ctrl+R` - Reverse search through history

clear (Clear Terminal)

```
clear
```

Tab Completion

- Press `Tab` to autocomplete commands, files, directories
- Works when there's a unique match

alias (Create Shortcuts)

```
alias foobar='ls -la'          # Create alias
unalias foobar                 # Remove alias
```

Permanent Aliases:

- Add to `~/.bashrc` file for persistence after reboot

7. Exit Commands

```
exit          # Exit shell
logout        # Alternative exit
```

Key Concepts to Remember

1. Everything in Linux is a file
2. File hierarchy starts from root directory (/)
3. Hidden files start with .
4. Use flags to modify command behavior
5. Tab completion saves time
6. Be careful with `rm` - deletions are permanent
7. Linux filenames don't need extensions
8. Commands can be combined with multiple flags

Common Flag Patterns

- `-a` - All/Show hidden
- `-l` - Long format/List details
- `-r` - Recursive (for directories)
- `-i` - Interactive (prompt before action)
- `-f` - Force (no prompts)
- `-p` - Parent/Create parent directories